CHAPTER FIVE

The Role of Spatial Statistics in the Control and Elimination of Neglected Tropical Diseases in Sub-Saharan Africa: A Focus on Human African Trypanosomiasis, Schistosomiasis and Lymphatic Filariasis

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Advances in Parasitology, Volume 97
ISSN 0065-308X
http://dx.doi.org/10.1016/bs.apar.2017.01.001

Abstract

Disease control and elimination programmes can benefit greatly from accurate information on the spatial variability of disease risk, particularly when risk is highly spatially heterogeneous. Due to advances in statistical methodology, coupled with the increased availability of geospatial technology, this information is becoming increasingly accessible. In this chapter we describe recent advancements in spatial methods associated with the analysis of disease data measured at the point-level and demonstrate their application to the control and elimination of neglected tropical diseases (NTDs). We further provide information on spatially referenced data sources and software that can be used to create NTD risk maps, concentrating on those that can be freely obtained. Examples relating to three NTDs affecting populations in sub-Saharan Africa are presented throughout the chapter, i.e., human African trypanosomiasis, schistosomiasis and lymphatic filariasis. These three diseases, with differing routes of transmission, control methods and level of spatial heterogeneity, demonstrate the flexibility and applicability of the methods described.

1. INTRODUCTION

In recent years we have seen great progress being made towards the improvement of human health across the developing world, with notable examples in sub-Saharan Africa including the halving of the number of clinical cases of malaria attributed to *Plasmodium falciparum* between 2000 and 2015 (Bhatt et al., 2015), and the near-eradication of Guinea worm with just 25 reported cases in 2016 (WHO Online Report, 2016). Neglected tropical diseases (NTDs) in particular have received a substantial amount of attention, with the World Health Organization targeting 8 of the 17 NTDs for elimination, in addition to the eradication of Guinea worm, i.e., onchocerciasis (in the Americas), lymphatic filariasis (LF), trachoma, leprosy, Chagas (in the Americas), visceral leishmaniasis (in Southeast Asia), Yaws (in Southeast Asia) and human African trypanosomiasis (HAT), whereas others such as schistosomiasis and soil-transmitted helminths are targeted for control (World Health Organization, 2010). Tools for control and elimination recommended by WHO include preventive chemotherapy, intensified case management, vector control and the provision of safe water, sanitation and hygiene (Hotez et al., 2010; World Health Organization, 2012; Freeman et al., 2013; Bockarie et al., 2013; Golding et al., 2015; World Health Organization, 2015). With many of these strategies in place, the geographical disease landscape is changing and will continue to change as the prevalence of these diseases declines. This decline leads to many operational challenges when designing and implementing control and elimination interventions, as there is an inverse relationship between a diseases level of endemicity and the efforts required to locate those with the disease. As a consequence, the strategies used to control diseases that are currently relatively commonplace will need to be adapted to ensure that they are feasible one the disease becomes rare, else resurgence may occur (Bergquist et al., 2015; Bockarie et al., 2013; Hawkins et al., 2016). To achieve this, a deeper understanding of where a disease is likely to persist is a very valuable resource. This chapter therefore aims to present spatial statistical methodology and spatial resources currently available that may assist in understanding the spatial distribution of NTDs. To demonstrate these advances, their applications to three NTDs found in sub-Saharan Africa are presented: HAT, schistosomiasis and LF (Lutumba et al., 2016; Danso-Appiah, 2016; Sodahlon et al., 2016). These three diseases vary in spatial heterogeneity (HAT is most heterogeneous, LF is the least), transmission routes, and approaches to control/elimination and as such serve to prove the diversity of the methods and resources being described in this chapter (see Table 1). In Section 2 we provide an overview of these three NTDs, and intersperse each subsequent section with examples of the statistical methods and resources being applied to further our understanding of their distributions, and potentially guide intervention strategies for controlling and eliminating them. Section 3 focuses on statistical methods that are commonly applied to disease risk mapping, and Section 4 summarizes common issues that are encountered when undertaking spatial analyses. Finally, Section 5 introduces a range of spatial data resources that are available to aid in mapping the risk of the three NTDs of interest.

Table 1 Environmental influences on the spatial distribution of human African trypanosomiasis, schistosomiasis, lymphatic filariasis in sub-Saharan Africa (SSA)

Disease	Parasite	Vector/ intermediate host	SSA burden	Primary methods of control	Environmental influences	Key environmental references
Human African trypanosomiasis (HAT)	*Trypanosoma brucei* gambiense, *Trypanosoma brucei* rhodesiense	Tsetse flies (*Glossina* spp.)	3796 cases reported in 2014	Case detection, vector control and control of animal reservoirs	Presence of rivers, vegetation, land cover, temperature, elevation, rainfall, proximity of livestock	Rogers (2000); Sciarretta et al. (2005); Batchelor et al. (2009); Wardrop et al. (2010); Matawa et al. (2013); Franco et al. (2014) and Dicko et al. (2014)
Schistosomiasis	*Schistosoma haematobium, Schistosoma mansoni*	Freshwater snails (*Biomphalaria, Bulinus*)	~192 million cases, estimated in 2003	Preventive chemotherapy	Presence of water bodies, vegetation, land cover, temperature, elevation, rainfall, soil properties, water properties, water accessibility	Lai et al. (2015) and Manyangadze et al. (2015)
Lymphatic filariasis	*Wuchereria bancrofti*	*Anopheles* and *Culex* mosquitoes	~40 million cases, estimated in 2013	Preventive chemotherapy	Presence of water bodies (transient and permanent), vegetation, land cover, temperature, elevation, rainfall, humidity	Slater and Michael (2013); Cano et al. (2014); Moraga et al. (2015) and Simonsen and Mwakitalu (2013)

2. OVERVIEW OF NEGLECTED TROPICAL DISEASES

NTDs are a diverse group of diseases, with their commonality being that they have traditionally been a low priority in those countries affected by them, despite their large, but potentially hidden, adverse impact on the population. Key features of the three geographically overlapping selected diseases are described further, including how they are transmitted, the factors influencing their spatial distribution, and strategies used for their control (Table 1). References which explore the spatial distribution of these diseases are included in this section, with more detail on the methods used being presented in Section 3.

2.1 Human African trypanosomiasis

HAT, also known as sleeping sickness, is an infection caused by protozoan parasites belonging to the Trypanosoma species. It is a vector-borne disease, with the vector being the tsetse fly, i.e., a fly of the *Glossina* genus. There are two forms of HAT depending on the parasite involved, i.e., *Trypanosoma brucei gambiense,* also known as Gambian HAT, which mainly occurs in West and Central Africa and accounts for the majority of HAT cases and *Trypanosoma brucei rhodesiense*, also known as Rhodesian HAT, which mainly occurs in East and Southern Africa. There are two stages of the disease. Symptoms of stage one are relatively mild and include fever, headaches and joint pain, whereas stage two is much more obvious, causing neurological conditions such as convulsions, confusion, sleep disturbance and eventual death if untreated. *T.b. gambiense* infections result in the chronic form of the disease, i.e., a long first stage, whereas those infected with *T.b. rhodesiense* experience neurological signs and symptoms more rapidly (World Health Organization, 2013a; Franco et al., 2014).

2.1.1 Geographical influences of transmission

HAT is the most spatially heterogeneous (focal) of the three diseases being considered and currently is the closest to elimination with 3796 cases reported in 2014 (World Health Organization, 2016). Approximately 360 HAT foci have previously been identified (300 for Gambian HAT, 60 for Rhodesian HAT), which are generally found in rural and remote areas. Tsetse flies are considered to be very sensitive to environmental conditions, hence even with these defined foci there is heterogeneity in disease risk (Fèvre et al., 2006; World Health Organization, 2013a). When determining the geographical distribution of disease prevalence, or more importantly the geographical distribution of transmission, the associated environmental factors depend on the *Glossina* subgenus and the form of disease under consideration. For example, Gambian HAT is primarily transmitted by riverine species of tsetse, hence transmission risk is strongly associated with the presence of rivers and the surrounding riparian vegetation (Rogers, 2000; Franco et al., 2014). Rhodesian HAT is primarily transmitted by savannah species hence is more associated with forests. Further, as the Rhodesian form is a zoonosis for which animals are the main reservoir, the presence of wild fauna and cattle also influence the distribution of risk (Batchelor et al., 2009; Wardrop et al., 2010). Factors also influencing a person's likelihood of coming into contact with infected tsetse also need to be considered.

Spatial models relating to the transmission of HAT either focus on the distribution of the vector or HAT cases. There are only a small number of papers in the published literature on Gambian HAT cases that incorporate spatial analysis, with the majority of these being produced by the Atlas of HAT team (Simarro et al., 2011, 2015; Lumbala et al., 2015). To our knowledge there are no papers that explore the link between the environment and Gambian HAT cases. For Rhodesian HAT, there are several papers that explore the relationship between cases and variables such as vegetation, temperature, elevation and the proximity of livestock to produce predictive maps of prevalence (Batchelor et al., 2009; Wardrop et al., 2010). More prolific in the literature are studies that explore the spatial distribution of the tsetse fly vector. Early examples of this are summarized by Rogers (2000), with more recent work including Sciarretta et al. (2005, 2010), Matawa et al. (2013) and Dicko et al. (2014). Common tsetse risk factors incorporated into these models include elevation, vegetation, temperature, rainfall and land cover.

2.1.2 Control and elimination strategy

Strategies for controlling HAT include the following:

- active case detection and subsequent treatment,
- passive case detection and subsequent treatment,
- vector control and
- control of animal reservoirs

or combinations thereof. In areas where humans are the main reservoir, the primary method of control is the detection and treatment of infected individuals. Active surveillance firstly involves screening the population by checking for clinical sign and performing serological tests (available for the Gambian form only). Diagnosis is then confirmed by establishing whether parasites are present in body fluids. If this is the case, a lumbar puncture is performed to determine the stage of the disease, i.e., to determine if the parasite has passed into the central nervous system, the results of which determine which treatment needs to be administered. Current treatment options are however considered to be toxic and complicated to administer (Kennedy, 2013). This active approach has been shown to be effective in some Gambian HAT areas, but in other areas where disease transmission continues to persist, vector control is also needed using approaches such as artificial bait methods (traps and targets), ground and aerial spraying, plus sterile insect techniques (World Health Organization, 2013a). It has been shown that reductions in the tsetse population of 72% or more can be enough to prevent transmission in areas where riverine tsetse are the main vectors (Gouteux and Artzrouni, 1996; Solano et al., 2013; Tirados et al., 2015). In Rhodesian HAT areas, animal reservoirs need to be considered. Where domestic livestock is the primary reservoir, treatment with chemotherapy or livestock-targeted vector control can be considered, whereas if wild animals are the primary reservoir, vector control and area avoidance are the only viable options available.

2.2 Schistosomiasis

In sub-Saharan Africa the two forms of schistosomiasis, urogenital schistosomiasis and intestinal schistosomiasis are caused by *Schistosoma haematobium* and *Schistosoma mansoni* respectively. Schistosomiasis is a waterborne disease such that infection occurs when people come into contact with freshwater inhabited by snails (*Bulinus* and *Biomphalaria* spp.) carrying parasitic blood fluke (Fenwick et al., 2006; Colley et al., 2014). More specifically, people are infected by the larval forms of parasite which swim in freshwater after emerging from the snail carrier and penetrate the individual's skin. Symptoms of schistosomiasis are dependent on the form that the individual infected with but include blood in urine, plus kidney damage, fibrosis of the bladder, bladder cancer and infertility in advanced cases (urogenital) or blood in stool and stomach pain, plus enlarged liver and spleen in advanced cases (intestinal form). Further, prolonged infection may result in anaemia, malnutrition, stunted growth and cognitive impairment (Colley et al., 2014; Danso-Appiah, 2016).

2.2.1 Geographical influences of transmission

Forty sub-Saharan African countries are considered endemic for schistosomiasis, accounting for 93% of the disease burden with an estimated 192 million cases out of 207 million globally (Steinmann et al., 2006; Adenowo et al., 2015). Transmission of schistosomiasis occurs in areas where people have contact with waterbodies which are a good habitat for the snail host through activities such as bathing, swimming, fishing and domestic tasks such as cooking, resulting in a very spatially heterogeneous disease landscape. In sub-Saharan Africa disease prevalence is still very high, although it is much closer to elimination in other regions with notable successes in interrupting transmission occurring in countries such as China and Brazil (World Health Organization, 2013d). Environmental factors to consider when determining the risk of disease transmission therefore include those associated with snail habitat suitability, e.g., precipitation, temperature, various soil properties, vegetation indices, land cover type and water properties, plus those associated with human water contact, e.g., water accessibility (Manyangadze et al., 2015; Lai et al., 2015).

A 2015 review of spatial models for schistosomiasis (Manyangadze et al., 2015) identified 36 papers published between the period 2009–13 which focus on the spatial distribution of disease transmission risk in Africa supplementing a review published in 2009 which focused on the period 1996–2008 (Simoonga et al., 2009). In contrast to HAT, the majority of these studies focus on the distribution of cases as opposed to the intermediate snail host. Of these, prevalence of disease as opposed to infection of intensity is of greater interested, likely due to the control strategy for the disease.

2.2.2 Control and elimination strategy

Whereas the other two diseases under consideration are targeted for elimination, the goal set by the WHO is to control the disease by ensuring at least 75% of school-aged children living in at-risk areas have access to preventive chemotherapy with praziquantel. Currently, praziquantel is delivered either annually or biennially to school-aged children in areas considered to be at high or moderate risk, respectively. Risk is assessed using prevalence surveys such that high risk areas are those with an estimated schistosomiasis prevalence (using parasitological diagnostic methods) above 50%, whereas moderate areas are those with prevalence between 10% and 50% (World Health Organization, 2006). In recognizing the diseases spatially heterogeneous distribution due to its dependence on freshwater sources, the WHO prevalence survey guidelines state ‘you need to survey specifically some areas that are near lakes, ponds, streams or irrigated areas… select a few schools close to the water and some a little further away'. On a practical level, however, praziquantel is often administered at the implementation unit level, usually a district, hence strategies are commonly based on the average prevalence for the implementation unit using a sample of 50 children per randomly selected school (World Health Organization, 2013c; Sousa-Figueiredo et al., 2015). Due to the known spatial heterogeneity of the disease, it has been suggested that this strategy will not enable eventual elimination to be achieved and more targeted approaches are required. Suggestions of suitable strategies include the use of lot quality assurance sampling, sampling based on a variogram analysis to measure the degree of spatial heterogeneity in the data and a strategy based on increasing the 'mapping resolution', i.e., the proportion of schools sampled (Sturrock et al., 2011; Sousa-Figueiredo et al., 2015). Snail control, although not commonly used, has also been identified as an effective historical method of reducing disease transmission that should be reconsidered by control programmes (Sokolow et al., 2016).

2.3 Lymphatic filariasis

LF is a vector-borne disease transmitted by mosquitoes. In sub-Saharan Africa, the disease is caused by the parasitic worm *Wuchereria bancrofti*, which is responsible for 90% of cases globally, and is primarily transmitted by *Anopheles* mosquitoes in rural areas and *Culex* mosquitoes in urban and periurban areas. Whilst a large proportion of infected people are asymptomatic, LF has a wide range of debilitating clinical manifestations the most common of which are lymphoedema, hydrocoele and episodic adenolymphangitis (Hotez and Kamath, 2009; Nutman, 2013).

2.3.1 Geographical influences of transmission

In sub-Saharan Africa 36 countries are considered endemic for lymphatic filariasis, with an estimated 40 million cases accounting for 65% of the global burden in 2013 (Ramaiah and Ottesen, 2014). Transmission patterns in disease are largely associated with factors relating to the distribution of the vector, plus factors associated with human contact with mosquitoes. As *Anopheles* mosquitoes are also a major vector for malaria, a large amount of research has been undertaken to determine the risk factors associated with its distribution, i.e., altitude, rainfall, temperature, humidity, land cover type and vegetation indices (Slater and Michael, 2013; Cano et al., 2014; Moraga et al., 2015). With regards to human-mosquito contact, as *Anopheles* are mostly active between dusk and dawn, contact is determined by the use of personal protection methods such as bed nets and the proximity of the home to suitable *Anopheles* habitat. Less research has been undertaken into the factors affecting LF transmission in urban environments, with linkages being primarily identified with the locations of environmental infrastructure, e.g., proximity to sanitation facilities, sewerage and drainage, as *Culex* breeding sites tend to be situated in stagnant water collections (Simonsen and Mwakitalu, 2013).

As with schistosomiasis, the majority of papers which apply spatial methods to LF data focus on LF prevalence, with prevalence obtained using immuno-chromatographic test (ICT) cards being most readily available. In a recent paper, Moraga et al. (2015) collated all geographically referenced microfilariae (mf) and ICT prevalence data for sub-Saharan Africa prior to the commencement of preventive chemotherapy to produce regional maps of both prevalence measures. Similar prevalence maps based on historical or baseline survey data can be found at the regional national or subnational level for many sub-Saharan African countries (Gyapong et al., 2002; Slater and Michael, 2012, 2013; Stanton et al., 2013; Cano et al., 2014; Mwase et al., 2014).

As vector control for LF becomes a greater focus (World Health Organization, 2011; van den Berg et al., 2013), knowledge of the spatial distribution of LF vectors is also of importance. There is a vast amount of information available on the spatial distribution of *Anopheles* produced by the malaria community, e.g., the Malaria Atlas Project produces predictive species distribution maps produced using mosquito occurrence data (De'ath 2007; Sinka et al., 2010). Predictive maps of *Culex* distribution have also been derived by the VectorMap project (www.vectormap.org) (Moraga et al., 2015).

2.3.2 Control and elimination strategy

Similar to schistosomiasis, the primary elimination method for LF is annual preventive chemotherapy using ivermectin plus albendazole if onchocerciasis is coendemic, or diethylcarbamazine (DEC) plus albendazole otherwise (World Health Organization & Global Programme to Eliminate Lymphatic Filariasis, 2011). Implementation occurs at the implementation unit level which is primarily the district level. The average prevalence of adults (aged 15+) at the implementation unit level is used to determine whether to deliver preventive chemotherapy, using a decision threshold of 1% based on rapid diagnostic tests, i.e., ICTs for the *Wuchereria bancrofti* antigen. The average prevalence is determined using the results of 50–100 people in two villages that are considered most likely to have ongoing transmission. No guidance is provided on how to identify these villages, however. Initially suggested baseline mapping strategies included the Rapid Assessment of the Geographical Distribution of Filariasis (RAGFIL) approach, which advocated applying a grid-based sampling scheme to ensure sampled villages were approximately 50 km apart, and using the variogram (see Section 3.1.2) of the resulting prevalence estimates to estimate the spatial correlation parameters and subsequently produce the smoothed risk surface (Gyapong and Remme, 2001; Gyapong et al., 2002). This approach was not adopted by WHO, however (World Health Organization, 2000; World Health Organization & Global Programme to Eliminate Lymphatic Filariasis, 2011). In addition to preventive chemotherapy, vector control and morbidity management are also advocated (World Health Organization, 2011; World Health Organization, 2013b).

3. STATISTICAL METHODS FOR DISEASE RISK MAPPING

Disease risk maps come in a variety of forms depending on their purpose and crucially the availability of data. In its simplest form, disease maps are visual representations of disease risk (incidence, prevalence, intensity, vector abundance etc.) in geographical space, either at the area-level, e.g., using spatially discrete district boundaries, or point-level across an area of interest. Historically, area-level maps of disease risk are more common; however, due to advances in geographical and computational technology, point referenced data is now much more attainable. It is almost the natural progression, once given a visual representation of the disease landscape, to look for patterns in its spatial distribution and subsequently begin to hypothesize as to what influences of these patterns. An early and well-known instance of this took place during a cholera epidemic in Soho, London, in 1854. During this period, the now renowned Dr John Snow used a map of cholera deaths to identify the source of the epidemic, namely the Broad Street waterpump (Warner, 1996; McLeod, 2000). This led to John Snow confirming his hypothesis that cholera was transmitted via polluted drinking water as opposed to miasmata (bad air), which was the dominant theory at the time. Spatial statistical methods therefore exist so that we can go beyond hypothesizing whether spatial patterns in diseases exist, and what causes these spatial patterns to occur 'by eye', and instead allow us to draw more objective conclusions. This area of statistics is vast and continues to grow, therefore in this paper we focus on spatial statistical concepts and approaches involving point-level data only, i.e., data observed at a specific point in geographical space, as opposed to area-level spatial data or point process data (Cressie, 1993b). In particular we focus on methods which we find commonly applied to tropical disease data, including the three NTDs of interest. The methods described in this section have been separated into two categories, namely methods of assessing for evidence of spatial dependency and methods for mapping spatial variability.

3.1 Assessing for evidence of spatial dependency

Spatial dependency is the backbone of many spatial statistical models. The concept of spatial dependency, sometimes termed spatial autocorrelation, is that events that occur close together in geographical space are more similar than events that occur further apart, i.e., the first law of geography (Tobler, 1970). In a tropical disease context, spatial dependency is likely to be present as nearby locations share the environmental or socioeconomic conditions that are favourable for disease transmission, e.g., the environment is suitable habitat for the vector, or the lack of appropriate water or sanitation facilities result in human behaviour that increases the risk of transmission. By ignoring the presence of spatial dependency when determining a suitable statistical model for the data, there is a risk that the relationship between the disease and potential risk factors under investigation may be misrepresented or missed altogether (Legendre, 1993; Thomson et al., 1999; Dormann et al., 2007).

We describe two common methods for assessing spatial dependency found in the tropical disease literature: Moran's I and the variogram (Moran, 1950; Cressie, 1993a). Whilst both serve to identify whether there is a spatial pattern to geographically referenced data, it is only the latter that provides information on the data's spatial correlation structure. We assume that we are interested in the observed data $y_1, \ldots, y_n$ which are realizations of the spatially continuous process $Y(x)$ measured at locations $x_1, \ldots, x_n$ in geographical space. In practice, it is common to assess whether there is any evidence of spatial dependency once any known trends have been removed, e.g., assess the residuals obtained after fitting a generalized linear model, but for explanatory purposes we shall assume that $Y(x)$ is the process of interest.

3.1.1 Moran's I

Moran's I provides an overall assessment of whether values associated with specific geographical locations are clustered, random or dispersed (Moran, 1950). The (Global) Moran's I statistic is calculated as follows:

$$I = \frac{N}{\sum_i \sum_j w_{ij}} \frac{\sum_i \sum_j (Y_i - \overline{Y})(Y_j - \overline{Y})}{\sum_i (Y_i - \overline{Y})^2}$$

where w_{ij} is the spatial weight between the two points measured at location i and location j, N is the total number of observed locations and $\overline{Y}$ is the mean of Y. Spatial weight matrices can take many forms. Commonly $w_{ij} = 1$ if points i and j are neighbours, with examples of neighbours of i including locations within a fixed distance, or points that when ranked by distance are at most the kth point furthest away from i (K nearest neighbours). As the mean and the variance under the assumption of no spatial dependency are known, the index can be standardized and the resulting Z score can be compared to the Gaussian distribution. The statistic I lies between -1 and 1, and significant positive values indicate spatial clustering, whereas significant negative values indicate dispersion.

3.1.2 The variogram

The variogram is an important tool for exploring whether there is evidence of spatial dependency in geographically referenced data and for obtaining preliminary estimates of the parameters that represent the spatial correlation structure of the data, i.e., the covariance parameters. The theoretical variogram or semivariance of $Y(x)$ can be written as

$$V(x_i, x_j) = \frac{1}{2}\mathrm{Var}\{Y_i - Y_j\}$$

$V(\cdot)$ is sometimes referred to as the semivariance. To define this further, we initially assume that the spatially continuous data $Y(x)$ has a normal distribution, and it is simply a combination of the true underlying outcome $S(x)$, plus some random variability/measurement error/random $Z(x)$. It is commonly assumed that $S(x)$ is a stationary, isotropic Gaussian process with zero mean and covariance defined as $\mathrm{Cov}\{S(x_i), S(x_j)\} = \sigma^2\rho(u)$ where u is the Euclidean (straight-line) distance between x_i and x_j. The correlation function $\rho(\cdot)$ is represented by the Matérn family of spatial correlation functions (Matern, 1960; Guttorp and Gneiting, 2006), which have two parameters κ and ϕ. The κ parameter represents the smoothness of the function, whereas ϕ represents the rate at which the correlation between two points decreases with increasing distance. For example, the exponential correlation function $\rho(u) = exp\{-u/\phi\}$ is a special case of the Matérn family, with $\kappa = 0.5$. As Cov $\{S(x_i), S(x_j)\}$ depends only on the separation distance u between two points as opposed to the locations of the points themselves, we say that the process is stationary. Further as the separation distance is independent of direction, we say that the process is isotropic. We assume that $Z(x)$ are identically distributed, independent Gaussian random variables with mean 0, variance τ^2. In the spatial statistics literature, this is referred to as the nugget effect (Diggle and Ribeiro, 2007). Fig. 1 is a theoretical representation of the variogram using parameters $\sigma^2 = 0.8$, $\tau^2 = 0.2$, $\kappa = 0.5$ and $\phi = 0.4$. Note that the practical range represents the value of u when the semivariance is equal to 95% of the sill, i.e., $\tau^2 + \sigma^2$. We note that in the presence of spatial correlation, the values of the variogram are small for points close together, and increase as the distance between points increases indicating that close points are more similar (less variable) than distant points. At a sufficiently large separation distance there is negligible correlation between the points, and the variogram line levels off and tends asymptotically to the value of the sill. The value of the semivariance, $v(u)$ at close or coincident points (small u) represents the variance of the random noise/measurement error, τ^2. This is sometimes referred to as the nugget effect. The steepness of the initial slope (i.e., the rate at which the spatial correlation decays) is determined by ϕ and κ.

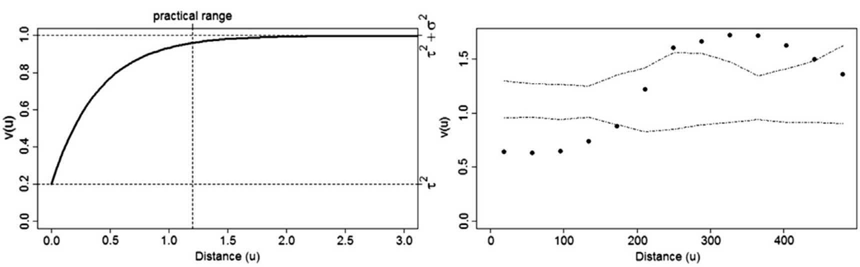

Figure 1 A depiction of the theoretical variogram with an exponential correlation function ($\kappa = 0.5$, with $\tau^2 = 0.2$ (nugget) and $\sigma^2 = 0.8$, and a practical range of 1.2 (left), and the empirical variogram of logit-transformed schistosomiasis prevalence data obtained for 299 schools in Namibia (Sousa-Figueiredo et al., 2015), indicating the presence of spatial correlation up to distances of 300 km. See Fig. 2 for a map of surveyed prevalence.

In practice, to make an initial assessment of whether spatial dependency is present, plus obtain initial estimates of the covariance parameters σ^2 and ϕ plus the nugget effect τ^2, the empirical variogram is plotted. Point pairs are grouped by distance bands $h(u) = (u - d,\ u + d)$, and $h(u)$ is plotted against the average of the empirical variogram ordinates $v_{ij} = \frac{1}{2}(y_i - y_j)^2$, where y_i and y_j are separated by a distance of $u \pm d$. The shape of the empirical variogram gives some indication of whether there is spatial dependency in the data, plus estimates of the model parameters can be made either subjectively, e.g., fitting a line to the resulting points by eye (see `eyefit` in the `R` package `geoR`), or using curving-fitting techniques such as least-squares (see `variofit` in `geoR`). This approach should be implemented cautiously, however, as the fitted variogram is dependent on the width of the distance bands selected, and multiple combinations of parameter estimates may result in equally well-fitting curves (Cressie, 1985; Diggle and Ribeiro, 2007). Whilst curve-fitting approaches are generally adopted within geographical information systems (GIS) software (e.g., within the Geostatistical Analyst tools in ArcGIS), within the statistical community these parameter estimates are more commonly used as initial values in optimization algorithms when fitting spatial regression models as outlined in Section 3.2.2. Monte Carlo approaches are used to assess whether the empirical variogram provides evidence of spatial correlation in the data, or whether the data from which it has been produced are spatially unstructured (Baddeley et al., 2014). In this context, the data are randomly allocated to the observation locations a large number of times (usually at least 99). For each simulation, the empirical variogram is computed and the maximum and minimum value for each value of the distance band are plotted to form a simulation envelope. This envelope represents the range of empirical variogram values that could feasibly be obtained under the assumption that the data are spatially independent. Hence, if the observed empirical variogram lies outside of this envelope, there is evidence of spatial dependency in the data. Fig. 1 displays the empirical variogram of logit-transformed schistosomiasis prevalence data from 299 schools in Namibia (Sousa-Figueiredo et al., 2015). This figure indicates that spatial correlation persists up to separation distances of 300 km.

Variograms have been used to describe the spatial dependency in HAT, schistosomiasis and LF in a number of scenarios. Most commonly, the variogram is used as a preliminary exploratory tool to indicate whether spatial dependency needs to be considered when undertaking a regression analysis to explore the spatial distribution of either disease cases or their vectors. Additional uses in the literature include the investigation of appropriate survey sampling strategies that account for spatial heterogeneity in schistosomiasis and LF cases (Gyapong and Remme, 2001; Sturrock et al., 2011) and to determine the appropriate placement of monitoring traps for tsetse (Sciarretta et al., 2005).

3.2 Mapping spatial variability

Knowledge of where tropical disease transmission is likely to occur and how many people it is likely to infect is very valuable, particularly when developing and implementing an intervention. Further, as disease programmes develop the technology or the resources to adopt more spatially targeted control strategies, information of disease risk at an increasingly fine spatial scale is becoming more sought after (Carter et al., 2000; Verity et al., 2014; Walz et al., 2015). In the following sections we describe two approaches for mapping spatial variability in disease risk, i.e., interpolation methods and spatial regression, specifically model-based geostatistics (Diggle and Ribeiro, 2007). Both approaches are used to predict the spatial distribution of disease at locations across geographical space, with the latter having the additional benefit of providing information on the causes of the geographical variability. These approaches make explicit use of the spatial correlation in the data. Other spatial mapping approaches commonly applied in tropical disease research which do not explicitly incorporate spatial dependency, but instead assume that all of the spatial variability in the data can be explained by observed covariates, are briefly described at the end of this section.

3.2.1 Spatial interpolation methods

Spatial interpolation methods fall into two categories: deterministic and stochastic (probabilistic). Deterministic methods are those which predictions of the value of $Y(x)$ are determined by the values of observed nearby points with no assessment of uncertainty being made, with a common method for this being weighted moving average approaches such as inverse distance weights (IDW). The IDW method assumes that the value of the variable of interest at an unsampled location is a weighted average of the surrounding sampled point values, with the weight of the contribution of each surrounding point being dependent on the distance between the sampled and unsampled locations. Hence if we are interested in making a prediction at location x, we assume

$$\hat{y}(x) = \frac{\sum_i w_i y(x_i)}{\sum w_i}$$

where w_i, the inverse distance weight, is simply the inverse of the distance between location x and location x_i to some power p, and usually $p = 1$ or 2. As well we choosing the form of w_i, the number of neighbouring points that contribute to the estimate also need to be considered. Popular options involve including a fixed number of nearest neighbour points, or including all points within a fixed radius (de Smith et al., 2015). Fig. 2 presents the inverse distance weighted schistosomiasis prevalence surface for northern Namibia on a 2.5 km by 2.5 km grid. In this instance, $p = 2$ and all points were considered when determining the inverse distance weight.

Kriging is a stochastic spatial interpolation method that was initially derived to aid those working in the mining industry to estimate the distribution of gold. It is from here which the term *geostatistics* originates (Krige, 1951; Matheron, 1976; Cressie, 1990). Kriging was initially described as a form of weighted averaging, i.e., $\hat{Y}(x) = \sum w_i(x) Y_i$, with the weights being selected to minimize the mean square prediction error in the resulting estimates, and in recent years it has been shown to fit into the spatial regression model framework (Cressie, 1990; Diggle and Ribeiro, 2007), incorporating the concept of a Gaussian process $S(x)$ as described earlier. There are a number of forms of kriging. For example, simple kriging refers to an approach which uses the mean of the observed data $(\bar{y})$ as an estimate of the true mean of Y (μ) when calculating the kriging weights, whereas ordinary kriging using the generalized least squares estimator $(\hat{\mu})$ to estimate the mean (Cressie, 1993b; Diggle et al., 1998). Universal kriging assumes that the true mean of Y is not a single value, but instead depends on covariates $d_k(x)$ such that $\mu(x) = \sum \beta_k d_k(x)$ where the covariates $d_k(x)$ may or may not be spatially varying (Goovaerts, 1997). In practice, universal kriging involves applying simple kriging to the residuals of the detrended data $(y' = y - \hat{\mu}(x))$, then adding these back on to the global trend. *Trans*-Gaussian kriging is the term used to describe the kriging process when the surface of interest itself is not Gaussian, but there exists a transformation such that the transformed data can be considered to be approximately Gaussian. For example, in spatial epidemiology we often deal with prevalence or count data, which do not follow a Gaussian

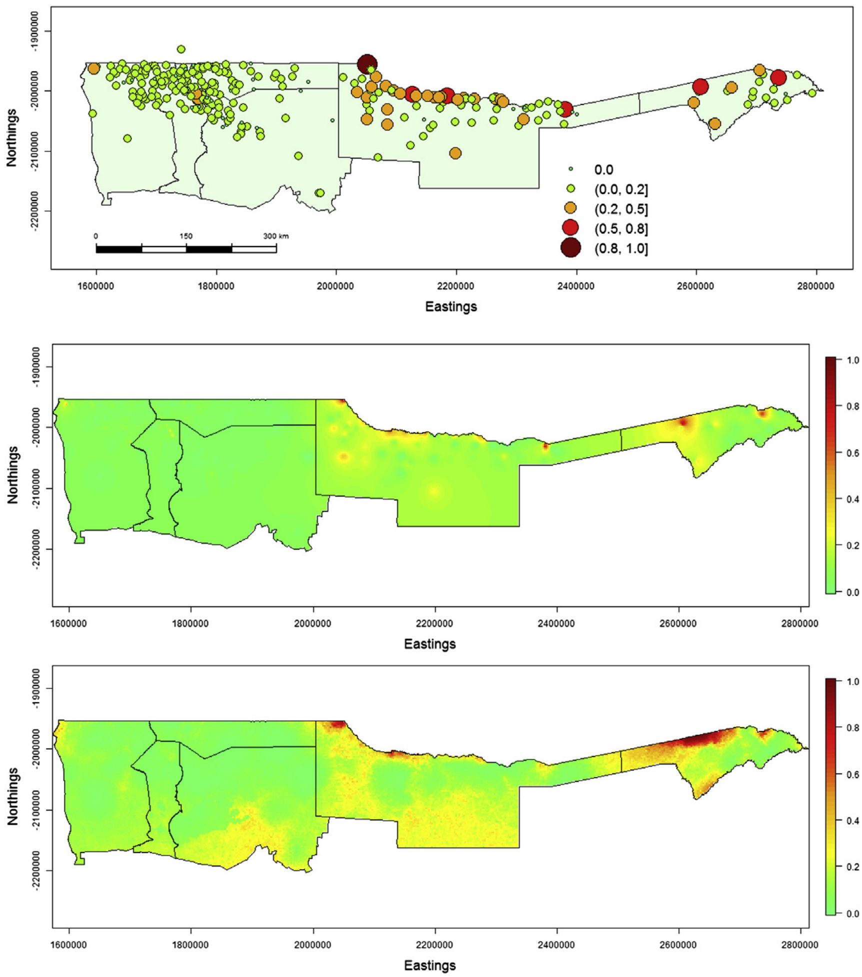

Figure 2 Surveyed schistosomiasis prevalence in Namibia (top), plus predictive map of schistosomiasis prevalence using inverse distance weights (middle) and the posterior mean schistosomiasis prevalence obtained from fitting a binomial geostatistical model to the prevalence data, including average maximum normalized difference vegetation index, average annual rainfall and topographical wetness index as covariates (bottom).

distribution. In this instance, transformations such as the *logit* transformation ($\text{logit}(x) = \log(x/(1-x))$ for prevalence data, or the log transform for count data need to be applied before kriging can be performed (Cressie, 1993b).

With the advent of new statistical methods that are able to directly model data from a range of distributions such as model-based geostatistics (see Section 3.2.2), 'traditional' kriging methods such as *trans*-Gaussian kriging has fallen out of favour with the statistical and disease research communities (Pullan et al., 2012), although due to its accessibility within GIS software such as ArcGIS, it is still frequently applied. Within the NTD literature, traditional kriging methods have been used to produce maps of tsetse fly abundance for HAT transmission risk (Sciarretta et al., 2005; Cano et al., 2007), schistosomiasis prevalence and snail abundance maps (Sturrock et al., 2011; Moser et al., 2014) and lymphatic filaraisis prevalence maps (Gyapong and Remme, 2001; Onapa et al., 2005; Koroma et al., 2012).

3.2.2 Spatial regression

Simple spatial interpolation methods such as IDW and simple and ordinary kriging are a valuable way of visualizing spatial trends in data; however, they rely upon the suitable selection of sampling points such that important changes in the outcome measure of interest may be omitted if there is heterogeneity in risk between sampled locations. Without information as to why the risk varies, these methods may result in these high ('hotspots') or low ('cold spots') risk areas being 'smoothed out', the consequences of which could be that those who are most at risk are excluded from control programmes. Spatial regression techniques incorporate spatial dependency into a regression framework, hence allow for an assessment of the effects of measureable factors on risk to be made without requiring the outcome of interest to be independent. As with generalized linear models, this information can then be used to predict risk at unsampled sites provided it is possible to obtain information on all of the covariates included in the regression model at these locations. In this chapter, we focus on the spatial regression method known as model-based geostatistics as this approach is popular in the tropical disease literature (Pullan et al., 2012).

Model-based geostatistics (Diggle and Ribeiro, 2007) embeds classical geostatistical techniques such as kriging into the generalized linear modelling framework. The benefit of this is that it relaxes the assumption that the outcomes of interest, or appropriate transformations thereof, must have a Gaussian distribution, and allows us to consider outcomes that follow a Bernoulli distribution (e.g., presence/absence data), a binomial distribution (e.g., prevalence data) and a Poisson distribution (e.g., incidence rates) as well as outcomes from other distributions in the exponential family. This extension of the geostatistical paradigm has led to model-based geostatistics being a popular method of assessing and subsequently mapping disease risk whilst accounting for the influence of (spatially varying) risk factors. For example, the binomial geostatistical model with the inclusion of risk factors is specified as:

$$Y(x)|S(x) \sim \mathrm{Bin}(m(x), P(x))$$
$$\mathrm{logit}\{P(x)\} = \sum \beta_k d_k(x) + S(x) + Z(x)$$

where $S(x)$ is a stationary Gaussian Process as described previously and $d_k(x)$, $k = 1, \ldots, K$ are risk factors which may or may not be spatially varying.

Parameter estimation within the generalized linear models paradigm is undertaken using likelihood-based methods. In simple terms, the optimal value of the parameters in the model are the values that optimize the likelihood function, i.e., a function which represents the chance of observing the data y given the parameter values. In extending the generalized linear model framework to include spatial correlation, maximum likelihood approaches become computationally complex (i.e., intractable), and subsequently alternative model-fitting approaches are required. This has led to the adoption of a Bayesian framework, such that rather than assuming the parameters of interest in the model are fixed (and unknown), it is assumed that they are random variables, and hence each has its own distribution. More details about Bayesian methods can be found in Gelman et al. (2013) and Diggle and Ribeiro (2007). Within the Bayesian framework, the standard approach for addressing the issue of intractable likelihood has been to use Markov Chain Monte Carlo (MCMC) methods (Gilks et al., 1995). Statistical software commonly used to fit generalized linear geostatistical models, notably WinBUGS (Lunn et al., 2000) and the `R` package `geoRglm` utilize MCMC to make inference, and over the last decade a plethora of research articles have been produced which apply these techniques to tropical diseases (Schur et al., 2011, 2013; Oluwole et al., 2015; Lai et al., 2015; O'Hanlon et al., 2016). These analytical developments have coincided with the increased availability of spatially referenced data on both the diseases themselves and their potential risk factors (See Section 4). However, unlike likelihood-based inference for generalized linear model, MCMC methods require the user to make numerous subjective inputs when specifying the model, which therefore requires the user having a reasonable working knowledge of the model-fitting process. Further, the model-fitting process can be computationally intensive, and relies upon the user's judgement to determine whether the MCMC algorithm has converged appropriately. As such, without sufficient knowledge and computing power, this modelling approach is considered to be difficult to perform.

In recognition of this, alternatives to MCMC have been sought, and the Integrated Nested Laplace approximation (INLA) algorithm is quickly being adopted as being a much less labour-intensive approach to fitting Bayesian models which contain a latent (i.e., unobserved) Gaussian process, such as geostatistical models (Rue et al., 2009; Lindgren et al., 2011; Lindgren and Rue, 2015; Brown, 2015). Whilst the user is still required to specify suitable initial values and tuning parameters, the model-fitting process is much quicker, and there is far less ambiguity with regards to assessing the reliability of the output. The R-INLA package (www.r-inla.org) provides an interface to INLA (a free-standing external programme), thereby enabling a wide range of spatial latent variable models to be fitted in `R` using the INLA methodology including model-based geostatistical models (Bivand et al., 2015). Due to the flexibility of R-INLA, the geostatistical model-fitting process is currently relatively difficult for a nonexpert, however. Software which acts as an intermediary between R-INLA and the less experienced geostatistical modeller is therefore under development, e.g., the package `geostatsp` (Brown, 2015).

The benefits of INLA over MCMC model-fitting approaches could potentially have an impact on the accessibility of these modelling approaches to those in resource-poor settings and further increase the feasibility of using the resulting predictive maps in a more dynamic way. For example, predictive maps could be updated with data collected in the field within a matter of hours, thus enabling control programmes to make decisions based on the most accurate spatial information of the disease to be available.

Whilst geostatistical models are not the only spatial regression approach available for disease mapping, they are useful as they not only result in predictive maps of disease risk, they provide several other key outputs that may be of interest to the disease control community. Firstly, as with all generalized linear models, in addition to a predicted risk surface, information on the uncertainty in predictions can also be easily extracted either in the form of prediction intervals, or perhaps more intuitively, exceedance probabilities, $\mathbb{P}(Y > T)$, where T is a threshold of interest. Secondly, models of this form not only provide information on which of the considered covariates are associated with the disease outcome, but further [assuming parsimony was taken into consideration when undertaking model selection (Lunn et al., 2000; Spiegelhalter et al., 2002; Austin and Tu, 2004; Hoeting et al., 2006)] provide information on the nature of those associations. Further, the estimated surface $\widehat{S}(x)$ indicates which geographical areas have a higher or lower disease risk after accounting for measured risk factors, which may aid further investigations to identify other unmeasured elements that may be affecting disease risk. Finally, these models are not restricted to only account for environmental variables as risk factors, and other influential measures, e.g., demographic, socioeconomic can be incorporated, with the caveat being that maps of the resulting risk can only be produced if the risk factor is available at each location where predictions of risk are to be made.

Fig. 2 presents the predicted mean schistosomiasis prevalence in Namibia, obtained by fitting a binomial geostatistical model using the R-INLA package. The model included the average maximum normalized difference vegetation index (NDVI) over a 3-year period obtained at 250 m resolution (NASA, 2016a), average annual rainfall at 100 m resolution (WorldClim, 2016) and topographical wetness index (TWI) derived from 90-m resolution elevation data (Sørensen et al., 2006; NASA, 2016b) as covariates. As with the map produced using IDW, the predictions were made on a 2.5 × 2.5 km grid. In comparison to the IDW map, the predicted prevalence is much more spatially variable as the relationship between the covariates and the prevalence survey data allows predictions to be made at locations far from those surveyed.

There are plenty of additional examples of model-based geostatistics in the schistosomiasis literature including large-scale schistosomiasis prevalence mapping (Schur et al., 2013; Lai et al., 2015), and intensity of infection in East Africa (Clements et al., 2006), plus the distribution of freshwater snails in Lake Victoria (Standley et al., 2012). Model-based geostatistical application to LF for sub-Saharan Africa are less common, with just a small number of papers using this approach to map LF prevalence at the large geographical scale (Slater and Michael, 2013; Moraga et al., 2015), whereas (Kelly-Hope et al., 2006; Stensgaard et al., 2011) use model-based geostatistics to explore the spatial association between LF and malaria. Similarly, there are currently few applications of model-based geostatistics to HAT transmission data, with publications predominantly focussing on Rhodesian HAT (Batchelor et al., 2009; Wardrop et al., 2010). As model-based geostatistics becomes more accessible through the development of more user-friendly software, it is possible that the popularity of this method will increase.

3.2.3 Common spatially implicit methods

There are numerous other disease risk-mapping approaches that do not explicitly incorporate spatial dependence, and instead assume that all of the spatially structured variability can be explained by measurable risk factors, commonly environmental variables. These methods often fall under the category of species distribution models or ecological/environmental niche models. Here we briefly describe several approaches that have been applied in the tropical disease literature, predominantly to map the spatial variability in disease vectors, i.e., MaxEnt, discriminant analysis and boosted regression trees. These methods acknowledge that the host–parasite relationship is very complex and nonlinear, hence intricate interactions need to be taken into consideration when trying to map disease risk. Although in theory it is possible to incorporate this complexity into spatial regression models such as geostatistical models, this would be at the expense of parsimony, and by extension, interpretability. These approaches generally make use of presence only or presence/absence data with the aim of predicting the probability that the species or disease of focus is present at a given location.

MaxEnt is a popular species distribution model as it only requires information on presence data to be able to elucidate the relationship between environmental variables and the species of interest (Phillips et al., 2006; Elith et al., 2006; Phillips and Dudík, 2008). This approach is commonly applied in ecology to determine the spatial patterns of the species using records of species sightings (i.e., presence only data), although it is also an alternative to logistic regression when there is doubt over the accuracy of absence data (Dicko et al., 2014). MaxEnt is a machine-learning technique, which in this context refers to automatic approaches which find the best combination (including nonlinear, complex combinations) of risk factors that describe the variability in the data whilst controlling for overfitting (Elith and Leathwick, 2009). This process is referred to as regularization. The MaxEnt software enables users to fit models of this form to their presence only data either using the software interface itself (https://www.cs.princeton.edu/~schapire/maxent/), or by using functions within the `R` package `dismo` (Merow et al., 2013; Hijmans et al., 2016). MaxEnt, and its accompanying software, is however a good demonstration of the disconnect between researcher working in different disciplines as it has since been demonstrated that MaxEnt and Poisson regression are in fact equivalent (Renner and Warton, 2013). By using regularization techniques such as lasso or ridge regression (Hoerl and Kennard, 1970; Tibshirani, 2011), users can fit the equivalent of MaxEnt models to their presence only data using `R` packages such as `glmnet` (Friedman et al., 2015), without restricting themselves to the limitations of the MaxEnt software.

With regards to the three NTDs of focus, MaxEnt has been used both in an ecological and an epidemiological context. For example, it has been used to provide species distribution maps for guiding tsetse control strategies (Matawa et al., 2013; Dicko et al., 2014), and for establishing the distribution of snails (Stensgaard et al., 2013; Pedersen et al., 2014), and has further been used to explore the large-scale spatial variability in LF (Slater and Michael, 2012; Mwase et al., 2014) using disease presence data at the regional and national level, respectively.

Popular binary classification methods that are frequently used to explore the spatial distribution of species, and perhaps less commonly, diseases, using both presence and absence data include discriminant analysis and decision trees. As with MaxEnt, the goal of these classification methods is to determine which combination of covariates best explains the data without focussing on the interpretability or parsimony. Nonlinear discriminant analysis focuses on identifying a combination of continuous covariates to discriminate locations where the outcome of interest is likely to be present or absent, and it has been used to map the distribution of disease vectors including mosquitoes, sandfly and tsetse fly (Robinson, 2000; Rogers, 2000, 2006). As the relationship between the outcome and the environment may be inconsistent across the full range of the environmental covariates under consideration, in many applications the environmental is stratified into areas of similar conditions. Nonlinear discriminant analysis is then undertaken within each of the strata separately, such that within each strata there is a nonlinear axis discriminating between the two groups in multivariate space (Rogers, 2000). Nonlinear discriminant analysis is equivalent to maximum likelihood classification, which is frequently implemented in GIS software to classify remotely sensed (RS) data; however, machine learning–based techniques such as decision trees (e.g., random forests) and support vector machines have been shown to consistently outperform this approach (Lu and Weng, 2007; Otukei and Blaschke, 2010; Cianci et al., 2015).

The boosted regression trees approach is an increasingly popular method being used in species distribution modelling and is a combination of two methods: regression trees and boosting (De'ath 2007; Elith et al., 2008; Stevens and Pfeiffer, 2011). Regression trees are a method of classification which aims to predict a continuous outcome. There are two stages to the process. Firstly the data are partitioned into small groups using a recursive partitioning approach. In brief, covariate-based binary decision rules are used to successively partition the data into smaller groups that are relatively homogeneous with regards to their relationship to the outcome. Once these groups have been established, simple regression models are then fitted to the data in each group independently (De'ath & Fabricius, 2000). Boosting is a method of improving the accuracy of a model and is based on the concept that it is easier to build multiple models based on a less strictly accurate decision rules and take an average that it is to build a single model based on a highly accurate ones. Boosting is therefore a sequential process where models such as regression trees are fitted iteratively to the data, and at each iteration the emphasis shifts to improve those outcomes that were poorly predicted in the previous iteration (Schapire, 2003; Elith et al., 2008). The boosted regression tree approach has been implemented in a number of recent prominent tropical disease epidemiology papers including an assessment of the global burden of dengue (Bhatt et al., 2013), and the global map of dominant malaria vectors (Sinka et al., 2010; Sinka et al., 2012), the global distribution of LF (Cano et al., 2014). Each of these examples utilizes presence and absence data rather than prevalence or abundance, with model-based geostatistics being the method of choice when suitable prevalence or abundance data are available (Hay et al., 2013b; Bhatt et al., 2015; Moraga et al., 2015; Grimes and Templeton, 2015).

4. COMMON ISSUES IN SPATIAL ANALYSIS

Spatial analyses of point-level data are sensitive to a number of factors which need to be accounted for when collating, analysis and interpreting the data. In this section we highlight several of the most common issues affecting tropical disease applications, i.e., the spatial scale at which the data are recorded, the sampling methods used to collect the data and the problems associated with combining data from different sources.

4.1 Spatial scale

In this context, when we refer to spatial scale we are predominantly concerned with the scale at which we would like to make our spatial predictions. This scale may be influenced by the objectives of the analysis, the characteristics of the disease under consideration, and the spatial scale of the available covariates. For example, the objective of the study may be to map the spatial distribution of disease at the large spatial scale (national, regional level) for descriptive purposes or to provide estimates of global burden, or maps may be required at a more detailed spatial scale to guide more targeted control interventions. The spatial scale of the former is therefore likely to be coarser than that of the latter. Consider for example a highly spatially heterogeneous disease such as HAT (see Section 1 for more details). Whilst predictions at a relatively coarse spatial scale are useful in providing a general sense of the distribution of disease (Simarro et al., 2010), due to the highly focal nature of the disease, important spatial variability in disease distribution will be masked at this scale and hence may produce inaccurate estimates of disease burden, or insufficient information to guide targeted intervention strategies (Sciarretta et al., 2005; Hackett et al., 2014). However, if disease risk is relatively spatially homogeneous within some geographical limit, for example, soil-transmitted helminths, it may be a waste of resources to produce maps at the finer spatial level (Sturrock et al., 2010).

Covariates used in the spatial analyses of tropical disease data are often derived from RS data (see Section 5). When incorporating these covariates, it is important to recognize that these data are already a spatial aggregation of a spatially continuous phenomenon, with the initial size and positioning of the boundaries of each RS cell (referred to in the statistical literature as the *support*) being defined by the instrument being used and the organization from which the data were sourced. Further, when undertaking the spatial analysis, these cells may be aggregated or disaggregated to ensure a consistent (and scientifically relevant) spatial resolution and alignment across all of the RS data being included in the analysis. The method by which the RS data are processed to be incorporated into the analysis can have a significant impact on the form of the relationship between RS data and the disease-related point-level data, and as such needs to be taken into consideration when evaluating the output (Atkinson and Graham, 2006; Raj et al., 2013; Hamm et al., 2015). This is a known problem in spatial statistics, referred to as the modifiable areal unit problem (MAUP) (Jelinski and Wu, 1996; Cressie, 1996; Gotway and Young, 2002). As a consequence, researchers may find that they draw different and sometimes opposing conclusions about the influence of a spatially aggregated covariate and the disease-related outcome of interest.

Whilst the MAUP may cause inconsistencies in the relationship between covariates and the disease-related outcome, it is often the case that the relative influence of covariates naturally differs at different spatial scales. For example, in epidemiology studies the dominant disease drivers at the large geographical scale tend to be associated with the climate and environment e.g., rainfall, temperature, elevation whereas at the smaller scale, the sociodemographic characteristics tend to become more influential (Simoonga et al., 2009; Hamm et al., 2015). Similarly, in ecological studies, e.g., when exploring the spatial distribution of disease vectors, the dominant drivers at the large-scale drivers are similar to those found in epidemiological studies, whereas the environmental nuances, e.g., fragmentation of habitat, river flow speed and steepness of the landscape need to be taken into consideration at the smaller spatial scale (Guerrini et al., 2008; Jacob et al., 2013; Hardy et al., 2015; Mweempwa et al., 2015).

4.2 Spatial bias

4.2.1 Sources of spatial bias

It is often the case that spatially referenced data that are used to explore the spatial variability in disease risk have been collected for a different purpose, with the spatial analysis being an opportunistic addition. As a result, when undertaking a post hoc spatial analysis, it is essential that the analyst is aware of any potential sources of spatial bias and adapt their models accordingly (Wardrop et al., 2014). For example, when conducting a survey, the location of samples is often influenced by the accessibility of an area such that vector sampling sites may not include densely forested areas, or epidemiological surveys may not be conducted in the more remote, impoverished communities. If collating historical data, differing diagnostic tools with nonequivalent sensitivity and specificity may be used in different geographical areas, or only data considered to be of epidemiological/ecological interest might be reported, excluding those geographical areas where disease prevalence or vector abundance is found to be low. Further, areas may be preferentially sampled, i.e., targeted specifically because they are suspected to have high or low disease risk. If such biases are identified, it's important that these are accounted for in the subsequent spatial analysis (Phillips et al., 2009; Diggle et al., 2010; Fourcade et al., 2014; Grimes and Templeton, 2015; Giorgi et al., 2015; Diggle and Giorgi, 2015).

4.2.2 Spatial sampling design

When designing studies to explicitly explore spatial variability, the spatial sampling design needs to be carefully considered, which includes the consideration of where to sample, how many locations to sample and how much data to collect at each sampled location, e.g., how many schoolchildren to survey per sampled school. Common sampling strategies employed in epidemiological surveys generally do not explicitly consider the spatial variation in the disease distribution and instead are designed to provide population-level summaries. These are referred to as design-based strategies. For example, at the small geographical scale, commonly employed design-based approaches include simple random sampling (where a suitable sampling frame is available) or systematic sampling using techniques such as the 'spin the bottle' approach (Bostoen and Chalabi, 2006). At the larger scale, to make the data collection more logistically feasible, cluster sampling is frequently employed where firstly the clusters (commonly villages or schools) are selected, following which individuals or households within the selected clusters are sampled. Sample size calculations for these types of survey are based on obtaining a sufficiently precise estimate of an area-level summary measure, e.g., prevalence. Whilst these approaches provide unbiased estimates of overall prevalence, they are not optimal for determining the spatial variability in disease risk within the study area. For highly spatially heterogeneous diseases, this may result in the important geographical areas of disease transmission being missed. A simple approach to addressing this is to use stratified sampling where the strata are geographically contiguous areas, with random or cluster sampling being used within the strata. Often, these strata are selected using environmental features, and whilst this may result in reduced within-strata spatial heterogeneity, this approach may not enable spatial variability to be explored in continuous space. Further, neither random, cluster, nor stratified sampling strategies account for spatial dependency when determining sample size, as in a scenario where there is high spatial dependency, sampling two locations that are geographically close together may not provide much more information than if one point were sampled in that area. Thus, the effective sample size in areas with high spatial dependency is reduced (Griffith, 2005).

Model-based sampling approaches are those that aim to learn about the spatial correlation structure of the data and make predictions at unsampled locations, as opposed to determining a single value to represent an entire area such as a mean. Within the spatial statistical literature, the location of the sampled points, as opposed to the quantity, is generally more of a consideration when determining an optimum sampling design (Diggle and Lophaven, 2006; Wang et al., 2012; Evangelou and Zhu, 2012). A commonly applied model-based sampling strategy for the purpose of spatial prediction is based on sampling at regular intervals in one dimension (transect) or two dimensions (lattice) (Diggle and Ribeiro, 2007; Miller et al., 2013; Buckland et al., 2015). Numerous examples of this can be found in the ecological literature, whereas spatial sampling has been less of a focus in tropical disease epidemiology, with examples of its consideration being relatively sparse (Gyapong et al., 2002; Zouré et al., 2011). If the focus of the research is to gain a better understanding of the spatial correlation structure of the data, the lattice design should be supplemented to also include close pairs of points within the lattice to estimate the spatial dependency over short distances (Diggle and Ribeiro, 2007).

In practice, spatial sampling strategies for diseases in resource-poor settings are likely to be influenced by pragmatism, with the number of sampling locations, plus the number sampled at each individual location, being limited by time and resources as well as scientific rigour (Magalhães et al., 2011). As disease landscapes change, as a result of a change in environment, behaviour as well as due to the influence of control, more consideration needs to be given to sampling strategies that promote the production of dynamic as opposed to static disease risk maps, e.g., adaptive spatial sampling (Peyrard et al., 2013; Siegfried and Siegfried, 2014) is an approach by which sampling is undertaken sequentially such that results of the previous samples are used to guide the selection of the next.

4.3 Combining different resources

Due to the lack of large-scale, spatially dense data from a single survey, disease maps at the national or regional level are often produced using historical data from multiple sources. Combining data from historical surveys can be very challenging, as the data under consideration often includes that collected using a variety of survey designs, e.g., different sampling strategies, diagnostic tests, target populations etc. In failing to incorporate these data heterogeneities into the model, the resulting predicted risk surfaces may be misrepresentative of the true underlying risk (Grimes and Templeton, 2015; Giorgi et al., 2015; Diggle and Giorgi, 2015). For example, it may be more appropriate to split the data into smaller, more homogeneous datasets before undertaking the analysis, to derive additional covariates which attempt to measure these differences, or to extend the model to include a temporal component.

5. SOURCES OF SPATIALLY REFERENCED DATA

Spatially referenced data is available in two categories, i.e., vector (points, lines, polygons) and raster. Due to the prolific increase in geospatial technology and the drive for open access data, there is now an abundance of spatially referenced data that is relevant to disease control in developing countries both publically and commercially available, plus we have the tools available to easily generate data ourselves. In the following sections we describe sources of data applicable to tropical disease control, with a focus on the three NTDs of interest (see Table 2).

5.1 Disease transmission data

With the growing realization that open access to survey data can have wide public health benefits, many leading scientific journals and research funders now require researchers to make their data available publically and numerous online global disease databases are being developed (Flueckiger et al., 2015). These databases often collate, and if necessary georeference, available disease data obtained from sources such as the published literature or directly from the researchers involved in collecting the data.

5.1.1 Global disease databases

There are a growing number of initiatives which aim to collate all historical spatially referenced prevalence/individual case data at the global scale for a specific disease. Example of this include the Global Atlas of Helminth Infections (www.thiswormyworld.org) which includes historical and contemporary data for LF, schistosomiasis and soil-transmitted helminths (Brooker et al., 2010; Cano et al., 2014; Sime et al., 2014), the Global NTD database which focuses on schistosomiasis, and the Atlas of HAT (Simarro et al., 2010; Lumbala et al., 2015). Not all data presented on these sites are available to download; however, some data may be shared on request. NTD Mapper (www.ntdmap.org) is an NTD-specific Website, currently focused on trachoma, Loa loa, LF, onchocerciasis, schistosomiasis and STH. Data presented come from a variety of sources, including prevalence surveys and literature searches with varying geographical scales. A large proportion of the data presented are available for download.

Contemporary and historical disease data may also be accessible from unpublished sources. For example, Healthmap (www.healthmap.org), founded in 2006, is an online resource for informal disease surveillance data, collating information from sources such as ProMed Mail, WHO and

Table 2 Resources for spatially referenced data relating to the risk of human African trypanosomiasis, schistosomiasis, lymphatic filariasis. Note that this list is not designed to be exhaustive, and other resources are available

Resource	Description	Data availability	References
Disease case data			
Global Atlas of Helminth Infections	Point prevalence survey data for **schistosomiasis**, **LF** and soil-transmitted helminths	All point prevalence maps, plus the raw data for a subset of countries are available to download	http://www.thiswormyworld.org/ (Cano et al., 2014)
Global NTD database	Point prevalence survey data for **schistosomiasis**	All point prevalence data available. Registration required	http://www.gntd.org/ (Hürlimann et al., 2011)
Atlas of human African trypanosomiasis	Number of HAT cases since 2000 by village	All case maps are available, and case data is available on request	http://www.who.int/trypanosomiasis_african/country/foci_AFRO/ (Simarro et al., 2010)
NTD Mapper	Point prevalence survey data for **schistosomiasis, LF**, soil-transmitted helminths, onchocerciasis, Loa loa and trachoma	Schistosomiasis, LF, soil-transmitted helminths and trachoma prevalence data are available to download	http://www.ntdmap.org/ (Flueckiger et al., 2015)
Healthmap	Various diseases	Locations of reported cases, and the source of the report are available to visualize online	http://www.healthmap.org/
Vector data			
Atlas of tsetse and African animal trypanosomiasis	Point-level tsetse count data	Maps and raw data currently unavailable, although preliminary maps found in Cecchi et al., (2015)	Cecchi et al. (2014)
Malaria Atlas Project	Predicted probability of occurrence of Anopheles mosquitoes as a 5 × 5 km resolution	Maps plus raster data are available to download	http://www.map.ox.ac.uk/ (Sinka et al., 2010)

(*Continued*)

Table 2 Resources for spatially referenced data relating to the risk of human African trypanosomiasis, schistosomiasis, lymphatic filariasis. Note that this list is not designed to be exhaustive, and other resources are available—cont'd

Resource	Description	Data availability	References
Remotely sensed data			
Landsat	Landsat data is available at 30 m resolution and includes 11 spectral bands, which include the visible spectrum, infrared, near infrared, short-wave infrared and thermal infrared. The data collection period is 1972–present.	The raw data are available to download at 30 m resolution using the USGS Earth Explorer. Landsat data has also been used to create a number of land cover products including GlobeLand30	http://landsat.usgs.gov/ http://earthexplorer.usgs.gov/ (download raw data) http://www.globallandcover.com/ (Globeland30)
MODIS	MODIS (Moderate Resolution Imaging Spectroradiometer) is available at 250 m resolution (2 bands), 500 m (5 bands) and 1 km (29 bands). The data collection period is 1999–present.	The raw data are available at their respective resolutions from LAADS Web. Real-time data can also be viewed using World View. Numerous products derived from MODIS data are also available to download, including MODIS land cover and NDVI.	https://ladsweb.nascom.nasa.gov/data (download raw data) https://worldview.earthdata.nasa.gov/ (view real-time data) http://modis.gsfc.nasa.gov/data/dataprod/ (list of products)
AVHRR	AVHRR (Advanced Very High Resolution Radiometer) data is available at 1.1 km resolution and currently includes 5 spectral bands in the red, near infrared and thermal infrared portions of the electromagnetic spectrum. The data collection period is 1979–present.	The raw data, plus products derived from the raw data such as NDVI are available from Earth Explorer	http://earthexplorer.usgs.gov/ (download raw data and NDVI products)

ASTER (Advanced Spaceborne Thermal Emission and Reflection Radiometer)	ASTER (Advanced Spaceborne Thermal Emission and Reflection Radiometer) data is available at 15–90 m resolution and includes 14 spectral bands in the visible, near infrared, shortwave infrared and thermal infrared portions of the electromagnetic spectrum. The data collection period is February 2000–present	The raw data are currently available to download from multiple sources including NASA Reverb	https://asterweb.jpl.nasa.gov/ http://reverb.echo.nasa.gov/reverb (download data)
SPOT	SPOT (Satellite Pour l'Observation de la Terre) is a commercial satellite with a resolution of 6 m, measuring 4 spectral bands (visible spectrum and near infrared)	A small subset of data is available to download for free from the European Space Agency. Registration is required.	https://earth.esa.int/web/guest/data-access
RapidEye	RapidEye is a commercial satellite cluster with a resolution of 5 m, measuring 5 spectral bands (visible spectrum, red edge and near infrared)	A small subset of data is available to download for free from the European Space Agency. Registration is required.	https://earth.esa.int/web/guest/data-access
Open Aerial Map	Open Aerial Map is a set of tools for sharing and using openly licenced satellite and unmanned aerial vehicle imagery.	Data availability is currently limited, with low geographical coverage.	openaerialmap.org

(*Continued*)

Table 2 Resources for spatially referenced data relating to the risk of human African trypanosomiasis, schistosomiasis, lymphatic filariasis. Note that this list is not designed to be exhaustive, and other resources are available—cont'd

Resource	Description	Data availability	References
Google Earth	Google Earth displays imagery collated from a number of sources, particularly commercial satellite vendors. The majority of the globe is covered in at least a spatial resolution of 15 m, with some areas having a resolution of 0.15 m.	Imagery accessed using Google Earth can only be used for visualization purposes.	https://www.google.com/earth/
Other data			
OpenStreetMap	OpenStreetMap (OSM) is a community-driven mapping project, which provides open data on landscape features such as transport routes (e.g., roads, footpaths, railways), water features (e.g., rivers, lakes), buildings, amenities, land use categories and administrative boundaries.	All data within OSM is available to download using a number of ways, including the OSM Website download area. Planet.osm can be used to download the whole dataset and is updated once a week.	http://www.openstreetmap.org/ http://planet.osm.org/ (download entire dataset)
WorldPop	WorldPop provides 100 m resolution data on human population distributions, modelled from census, survey, satellite and GIS datasets, including OSM.	Estimated population data are available at 1 km for Africa, Asia and South and Central America, whereas 100 m data is available for a subset of countries within these continents.	http://www.worldpop.org.uk/

Google News to produce maps of reported cases of a variety of diseases. This resource cannot be considered to be a complete record of cases, but may allow geographical areas of interest to be identified.

5.1.2 Vector/intermediate-host databases

There is a wealth of vector distribution data in the published literature, and as with disease data, there are initiatives in place to collate this information and make it more easily accessible. There are however many gaps remaining in data relating to the three NTDs of focus. For example, the Food and Agriculture Organization of the United Nations (FAO) are developing an Atlas of tsetse, which in combination with the disease cases atlas (Atlas of HAT) is intended to provide comprehensive information on disease transmission risk to national control programmes (Cecchi et al., 2014, 2015). A similar project has been undertaken by the malaria control community to create maps of malaria parasite prevalence (Guerra et al., 2007). This tsetse data does not however appear to be publically available. Due to its linkages with malaria, there is an extensive amount of information available on the distribution of *Anopheles* which may be of interest to the LF elimination community, notably that produced by the Malaria Atlas Project, although this focuses on mapping presence of mosquitoes at the large geographical scale and may therefore be of limited operational use. Less has been done to develop resources relating to the distribution of *Culex* mosquitoes, despite being transmitters of LF and West Nile virus in sub-Saharan Africa. To our knowledge, there are no resources relating to the spatial distribution of schistosomiasis-transmitting snails available. It is also worth noting that it is rare to find spatial information relating to the infection status of the vectors/intermediate hosts for any of the diseases under consideration.

5.1.3 Spatial data collection

As well as using previously collected data, researchers may also want to collect their own spatially referenced data to analyse. The capacity for the public to record precise geographical location data has been available since May 2000, when the US government switched off the pseudorandom errors in the publically available GPS signal, known as 'selective availability'. Since this date it has become increasingly easy to record location information. Initially, dedicated GPS receivers were required to record coordinates during field activities, which were often prohibitively expensive. However, since the proliferation of the GPS-enabled smartphone and tablet computers, this is no longer the case. Further, unlike many dedicated GPS receivers, smartphones/tablets have many additional features that can be taken advantage of when collecting disease data in the field. For example, they could be used to collect survey responses which can be automatically uploaded to a database once a data connection is available, thus circumventing the need for manual data entry. There are an increasing number of products and services available to develop these data collection platforms, some of which are designed to be accessible to researchers who do not have any programming skills. One popular example of such a product is the open source, Android-based OpenDataKit (ODK, www.opendatakit.org) (Anokwa et al., 2009; King et al., 2013). ODK is a suite of tools that enable users to develop their own data collection forms that can incorporate text and numeric data, GPS coordinates, photographs, videos and barcodes, which can be used to collect field data without being connected to either the mobile network provider or the Internet. An Internet connection is only required to upload the data, with the data being uploaded to online servers that are either hosted by Google's App Engine, or hosted locally. ODK has been successfully used to collect tropical disease data (King et al., 2013; Sime et al., 2014; Tom-Aba et al., 2015; Fähnrich et al., 2015), the largest scale of which has been in relation to trachoma mapping (Pavluck et al., 2014), such that between 2012 and 2015 the Global Trachoma Mapping Project (GTMP, http://www.sightsavers.org/gtmp/) collected data from 2.6 million people across 29 countries using ODK. Whilst large-scale mapping projects tend to be led by dedicated survey teams, as smartphones become more ubiquitous in developing countries, there is the growing potential to collect spatially referenced health data directly from the affected communities (Stanton et al., 2016b). Community members can either be requested to submit data related to specific field activities, or data can be obtained on a more voluntary or passive basis. Participatory data collection approaches have the potential to provide vast amounts of real-time spatially referenced data that could be of relevance to disease control (Hay et al., 2013a). The most successful examples of the use of crowd-sourced spatial data to improve health outcomes in the developing world primarily relates to improving geographical maps to aid disaster response (Zook et al., 2010; Médecins Sans Frontières, 2014), and detecting epidemics (Broniatowski et al., 2014; Milinovich et al., 2015) although there are also examples in the literature of participatory mapping of malaria vector exposure (Fuller et al., 2014; Mwangungulu et al., 2016).

5.2 Population movement data

A limitation of spatial methods is that they make the implicit assumption that the subject of interest is associated with a single geographical location, e.g., survey participants are usually assigned the coordinates of their homes, schools or villages when undertaking prevalence surveys, when it is of course recognized that exposure to disease risk may not occur at that location. In a purely spatial model this limitation is often accounted for by considering environmental risk factors within a fixed sensible buffer of the assigned location, or asking participants questions that might identify activities that expose them to risk outside of their homes/schools/villages. These approaches cannot, however, capture all of the intricacies of the participant's movements that may be linked to disease risk and could weaken the spatial signal in the outcome of interest. There is therefore a growing interest in improving measures of human mobility both in terms of large-scale patterns such as migration and small scale day-to-day movement (Kraemer et al., 2015). One research area of growing interest is the harnessing of mobility information collected by mobile phone operators via anonymized call logs. These human mobility data have been shown to be complementary to travel survey data and provide a deeper understanding of the spatial dynamics of infectious diseases in developing countries over time (Buckee et al., 2013; Wesolowski et al., 2014, 2015; Flowminder Foundation, 2017). On a smaller geographical scale, GPS-enabled wearable technology, which in its most basic form consists of a small GPS data logger which records coordinates at regular time intervals, can provide detailed information on human movement that might be of relevance to disease control. For example, GPS data loggers have been used to measure differences in infected and uninfected mother and children's spatial movement patterns in relation to schistosomiasis (Stothard et al., 2011), and assess the impact of LF-related morbidity on mobility (Stanton et al., 2016a).

5.3 Geographical disease risk factor data

Here we restrict ourselves to consider physical measures of the natural and man-made environment as opposed to other risk factors that can be spatially referenced, e.g., socioeconomic measures associated with a particular location.

5.3.1 Natural geographical features

RS data refers to data acquired from a distance, with the term being most commonly associated with data obtained from satellites. In areas where in situ environmental data is sparse, such as sub-Saharan Africa, satellite-derived RS data are often the only environmental data available at a spatially and temporally consistent scale, hence are an exceptionally valuable resource. Whilst the exact number of operational satellites is unknown, approximately 940 spacecraft have been launched since October 1957 for the purpose of earth science (Belward and Skøien, 2014; NASA, 2016c). These satellites tend to operate at a relatively low altitude (~700–800 km) and provide information at a range of spatial resolutions. As the value of these data continues to be realized, RS datasets that were once only available at considerable financial cost are being released to researchers or the general public for free. The Landsat and ASTER programmes are good examples of this.

NASA's Landsat programme has been providing data on a global scale since its launch in 1972, and in 2008 decided to release all data to the public (Woodcock and Allen, 2008). Landsat 1 was equipped with a multispectral scanner and initially provided data in four spectral bands (red, green and two infrared) at a spatial resolution of 80 m, with data for each geographical location being provided every 18 days. Most recently, Landsat 8 was launched in 2013 and is equipped with two instruments, namely the Operational Land Imager and the Thermal Infrared Sensor which provide data on the visible spectrum (red, green, blue), plus infrared, near infrared and short-wave infrared at 30 m resolution, thermal infrared at 100 m resolution and a grayscale/panchromatic image is produced at a resolution of 15 m. Data are provided every 16 days and are available to download using the USGS Earth Explorer (http://earthexplorer.usgs.gov/) shortly after collection.

ASTER (Advanced Spaceborne Thermal Emission and Reflection Radiometer) is a Japanese owned sensor on board NASA's Terra satellite. It was launched in 1999 and is still providing data today, despite its initial 5-year design life. All ASTER imagery was made available to the public in April 2016, prior to which only digital topographical maps were available. ASTER data is comprised of 14 spectral bands: four bands in the visible and near infrared ranges (15 m resolution), six bands in the shortwave infrared ranges (30 m resolution) and five bands in the thermal infrared ranges (90 m) resolution (Abrams et al., 2015). As with Landsat, data are provided every 16 days and are available to download using the USGS Earth Explorer.

Products such as Landsat require the user to be well versed in remote sensing and/or GIS to extract information that may be of relevance to their disease of focus, e.g., Landsat data can be used to derive measures of vegetation such as the NDVI, or classification algorithms can be applied to identify land cover type. In acknowledgement of this limitation, there are an increasing number of preprocessed resources available to non-RS/GIS experts. With regards to land cover, for example, whilst users continue to have the opportunity to use RS data such as AVHRR (Advanced Very High Resolution Radiometer, ~1 km resolution), MODIS (Moderate Resolution Imaging Spectroradiometer, 250 m/km resolution) and Landsat to use image classification techniques to derive land cover categories of interest (Townshend et al., 1991; Friedl and Brodley, 1997), there are also a number of global-scale land cover products freely available. The disadvantage of using these products is that they provide land cover information for only one time point, which may not match that of the disease-related data under consideration. Examples of global land cover products include MODIS Land Cover (250 m resolution, annual data 2001–12; Friedl et al., 2002, 2010), GLC-SHARE (1 km, data collated between 1998 and 2012), GlobCover [300 m resolution, 2009 RS data (Congalton et al., 2014);] and GlobeLand30 (30 m resolution, 2010). These data have been used extensively in the development of predictive disease models and maps, including the NTDs of focus, e.g., tsetse mapping (Kitron et al., 1996; Odiit et al., 2006; Guerrini et al., 2008; DeVisser and Messina, 2009).

Elevation is an important factor when considering the distribution of many tropical diseases, as it often impacts transmission, e.g., through influencing vector habitat suitability. The highest resolution global, publically available elevation data is currently 30 m and was collected by the Shuttle Radar Topography Mission (SRTM) in February 2000. This data was made available to the public at its finest scale in 2015 and can be accessed using the USGS Earth Explorer (http://earthexplorer.usgs.gov). As well as elevation, this data can be used to derive information on river networks using hydrological modelling. For example, the HydroSHEDS (HYDROlogical data based on Shuttle Elevation Derivatives at multiple Scales) has produced river network and drainage basin datasets using 90 m resolution SRTM data (http://hydrosheds.cr.usgs.gov/).

Data from civilian earth observation satellite sensors provide a vast amount of contemporary information on the global environment; however, given that the multispectral spatial resolution of these sensors rarely exceeds 30 m, important small-scale details associated with disease risk may be masked (see Section 4). When considering disease risk at a reasonable large geographical scale, this loss of information may not be of importance. However, as interest in more targeted disease control efforts, more detailed environmental information may be required. Commercial earth observation satellites include RapidEye (6.5 m multi-spectral resolution), SPOT (6 m multispectral resolution), IKONOS (3.2 m multispectral resolution), QuickBird (2.62 m, decommissioned in 2015), Pleiades (2 m multispectral resolution), GeoEye (1.84 m multispectral resolution) and WorldView (1.24 multispectral resolution), and archived data can be purchased from the satellites operators (DigitialGlobe, GeoEye, Blackbridge etc.) or from various value added resellers. The price of these products is very variable, depending on the resolution of the image, the size of the area required and the time period it is required for, and as such it has not been extensively applied to NTD research (Hamm et al., 2015), although the potential of this resource is becoming increasingly recognized as mapping disease risk at the small (micro) spatial scale becomes increasingly important, particularly for diseases where targeted control is possible (Bousema et al., 2012; Jacob et al., 2013; Walz et al., 2015; Shaw et al., 2015; Rayaisse et al., 2015). There are a number of initiatives to release archived very high resolution data for free, however. For example, the European Space Agency (ESA, https://earth.esa.int) has made their own archives of SPOT and RapidEye data available to the public following a brief registration process, whereas other datasets (e.g., QuickBird, GeoEye, WorldView, PLEIADES) are available to researchers on the submission and acceptance of a research proposal. The ESA does not however store all data produced by these commercial satellites in their archives hence there are still gaps in both geographical space and time. A small amount of very high resolution data can also be obtained from Open Aerial Map (https://openaerialmap.org), which is a platform for sharing open-licenced RS imagery, although this resource is in its early stages of development.

Google Earth is another valuable resource for researchers as it enables the user to visualize high resolution products such as those listed earlier and to create simple vector features (points, lines, polygons). Whilst this is an excellent resource, it is limited in use as users only have access to data from a small subset of time points, the resolution at which data are available are not geographically consistent, and users are not able to access the raw data and subsequently any of the additional spectral bands (Lozano-Fuentes et al., 2008; Jacobson et al., 2015). Finally, unmanned aerial vehicles (UAVs), commonly known as drones, are also now recognized as an accessible method of generating high resolution RS data that is applicable to public health (Fornace et al., 2014).

5.3.2 Man-made geographical features

Whilst historically there is a disparity in the availability of accurate, digitized, georeferenced data on the man-made landscape (roads, settlements, health facilities etc.) between the developed and developing world, advances in geospatial technologies are reducing this knowledge gap. We focus on two types of data resources in this paper, namely voluntary geographic information (VGI) resources and RS resources.

VGI resources are geographical information resources that are compiled from geographically referenced data provided by volunteers. Other terms used for this are participatory mapping or crowdsourced mapping (Goodchild, 2007; Sui et al., 2012). An excellent example of a VGI resource on a global scale is OpenStreetMaps (OSM, www.openstreetmap.org) (Budhathoki and Haythornthwaite, 2012; Neis and Zielstra, 2014). OSM was founded in 2004 with the goal of creating a geographic database that was both free to use and contribute to, thus harnessing the power of local knowledge and the enthusiasm of experienced and amateur mappers. Data contributed to OSM can be that collected directly in the field using GPS-enabled devices or digitized from sources such as aerial images or locally generated paper-based maps. Vector data created by users are publically available via numerous resources such as Planet OSM (http://planet.openstreetmap.org/) or using the OSM import functionality within QGIS (http://wiki.openstreetmap.org/wiki/QGIS). See http://wiki.osmfoundation.org/wiki/How_To_Get_OpenStreetMap_Data a more comprehensive list of sources. The level of detail available within a geographical area of interest, and the quality of the data, is however dependent on the contributions made by the OSM community and hence is not globally consistent (Haklay, 2010; Neis et al., 2011; Arsanjani et al., 2013; Barron et al., 2014).

In recognition of the power of maps for humanitarian aid and economic development in the developing world, the Humanitarian OpenStreetMap Team (HOT) was established (www.hotosm.org). Through the medium of the HOT Task Manager, OSM users are asked to assist in improving the quality of maps in particular areas of interest with regards to humanitarian response. This was first implemented in response to the 2010 earthquake in Haiti, and more recently the HOT has been involved in mapping areas in West Africa affected by Ebola (Teng et al., 2014; Soden and Palen, 2014; Médecins Sans Frontières, 2014; Palen and Soden, 2015; Koch, 2015). Not only do these maps provide valuable information to those directly responding to these crises, but they leave behind a legacy of freely available geographical information that can be used to benefit these population further (Eckle and Albuquerque, 2015; Stevens and Pfeiffer, 2015). In addition to providing maps for immediate humanitarian response, HOT also supports community development. Current projects supported by HOT include mapping all residential areas, all building plus water and sanitation facilities southern parts of Malawi affected by extensive flooding in January 2015 to improve flood risk management in area and similarly mapping infrastructure relating to flood risk in Dar es Salaam, Tanzania following a series of flooding events over the last five years. These community development initiatives are part of the Missing Maps Project (www.missingmaps.org), which supports HOT through promoting the mapping of vulnerable populations preemptively rather than responsively. Fig. 3 presents a comparison between Google Maps and OSM of the historic downtown of Monrovia, Liberia, highlighting the level of detail available in OSM as a result of the Ebola crisis.

As these spatial data increase in accuracy, we speculate that older resources for man-made features will be replaced by more contemporary OSM data, which have the capacity to improve our knowledge on the spatial distribution of disease risk. For example, the population-mapping project WorldPop (http://www.worldpop.org.uk/) has started to integrate OSM data into their maps to improve the precision of their 100 m spatial resolution population density maps (Linard et al., 2014; Sorichetta et al., 2015). These population data can subsequently be used to obtain more accurate estimate of disease burden. An improved knowledge of infrastructure, and subsequently the spatial distribution of the human population, can also aid in understanding the distribution of disease risk at finer spatial scales through the derivation of relevant risk indices. For example, detailed land use and transport network data can assist in determining the accessibility of an area, or the proximity between an at-risk population and geographical risk factors. More practically, resources such as OSM can be used to better plan the distribution of interventions and treatments.

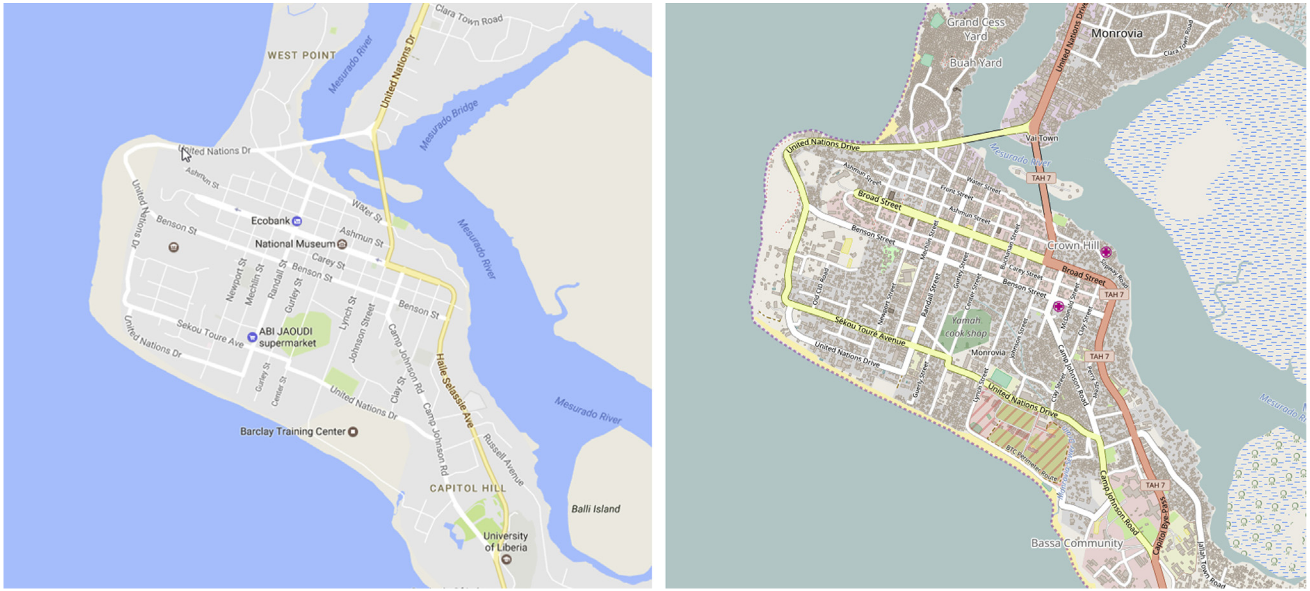

Figure 3 A comparison of the level of detail provided by Google Maps (left) and OpenStreetMap (right) for Monrovia, Liberia.

6. CONCLUSIONS

It is encouraging to see the quantity of spatially referenced data that are being made publically available, particularly in terms of disease prevalence and RS data. Not only is the spatial resolution of the available data becoming increasingly fine, some data, particularly RS data such as Landsat and ASTER are available at a continuous temporal resolution in near-real time. Thus detailed contemporary spatio-temporal models and maps are increasingly achievable. This trend needs to continue to allow researchers and control programme implementers to maximize on their potential to benefit disadvantaged populations. There are however still gaps in the data that need to be addressed to understand the changing epidemiology of the diseases of interest and increase the likelihood of sustained control and elimination. For example, more regularly collected geographically referenced disease monitoring and surveillance data is required at the spatial scale of interest to the respective control programmes. Further, and crucially, to avoid premature cessation of control efforts, more sensitive diagnostics, plus sustained vector/intermediate-host surveillance are required for all three diseases (Hawkins et al., 2016; Hotez et al., 2016; Stothard et al., 2017).

This chapter has demonstrated the utility of spatial methods in tropical epidemiology, showing that a great deal has been learnt about the spatial patterns of diseases targeted for control and elimination in recent years as a result. The focus should now be on putting the information acquired from spatial statistical methods into operational use (Bergquist et al., 2015). There have been some examples of this within the context of HAT control, a disease that is close to elimination and highly spatially heterogeneous, and hence engenders a more targeted approach (Sciarretta et al., 2005, 2010). Current control strategies for the other two diseases of focus treat the selected at-risk population with preventive chemotherapy indiscriminately at present. However, as prevalence decreases as a result of control a more targeted approach may be necessary, particularly if the decrease is not spatially uniform. It is therefore essential to consider how statistical models and spatially referenced data can be fully utilized to make the end stages of these programmes as resource efficient as possible. It is however worth bearing in mind that in situations where indiscriminate control strategies are much more operationally feasible than the costs associated with developing a targeted approach, that the former may be a more valid elimination strategy than the latter for some diseases

(Stothard et al., 2014). In developing risk maps to inform a public health intervention, it is always necessary to be mindful on the fine balance between producing risk maps that are incredibly detailed and accurate and producing maps that are operationally useful. Thus, the lines of communication between the data providers, the statistical modellers and the disease control implementers should always be open to maximize the potential impact of the end product and ultimately reduce the burden of disease.

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